

Data and Artificial Intelligence

Cyber Shujaa Program

Week 5 Assignment

Data Visualization on Tableau

Student Name: Ashley Adhiambo

Student ID: CS-DA03-26058

Table of Contents

Introduction	3
Tasks Completed	3
Step 1: Understanding the Business Requirements	3
Step 2: Loading and Transforming Data	4
Step 3: Creating Calculated Measures	5
Step 4: Designing Visualizations.....	5
Step 5: Building the Dashboard	7
Step 6: Detailed Employee Records View	7
Conclusion.....	8

Introduction

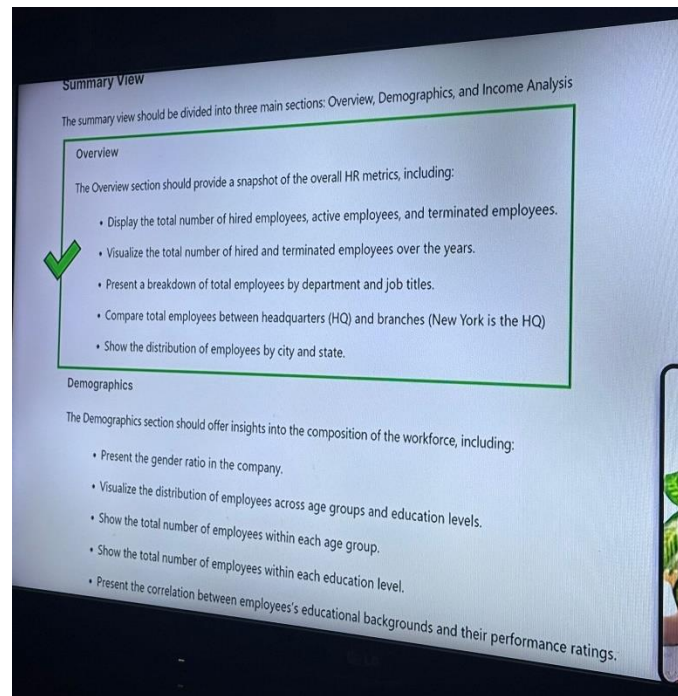
This assignment involved building an interactive Human Resources (HR) dashboard using Tableau Desktop for data-driven decision making. The goal was to simulate a real-world HR analytics project by understanding business requirements, preparing and transforming data, creating calculated measures, designing multiple visualizations, and assembling them into a clean, interactive dashboard.

The dataset used in this project was provided as part of the download materials. It contains simulated employee records, including demographics, job details, performance, and salary information. The HR manager's requirements as stated in the tutorial guided the overall design of the dashboard, which needed to be divided into three main sections: Overview, Demographics, and Income Analysis, as well as a detailed employee table.

Tasks Completed

Step 1: Understanding the Business Requirements

The initial step involved interpreting the HR manager's requirements to ensure that the dashboard would meet organizational needs. The dashboard was expected to provide a comprehensive overview of workforce metrics, such as the total number of employees, new hires, and terminations, as well as trends over time. The demographic breakdown was also crucial, covering gender ratios, age distribution, and education levels. Additionally, the dashboard needed to highlight salary comparisons and provide a detailed table of employee records that could be dynamically filtered to explore specific segments of the workforce. Understanding these requirements helped shape the structure and focus of both the visualizations and the overall dashboard layout.



Step 2: Loading and Transforming Data

The HR dataset was imported into Tableau, and initial data inspection ensured that field types were accurate. Dates, categorical fields, and numerical fields were standardized to enable proper analysis. During the transformation process, field names were clarified for readability and categories such as job titles and departments were harmonized. Geographic hierarchies were created by combining state and city information, which allowed for effective map visualizations. These preparation steps ensured the dataset was accurate, clean, and ready for analysis.

Tables	
Age Groups	
Birthdate	
Department, Job Title	
Department	
Job Title	
Education Level	
Employee ID	
First Name	
Full Name	
Gender	
Hiredate	
Last Name	
Performance Rating	
State, City	
Location	
State	
City	
Status	
Termdate	

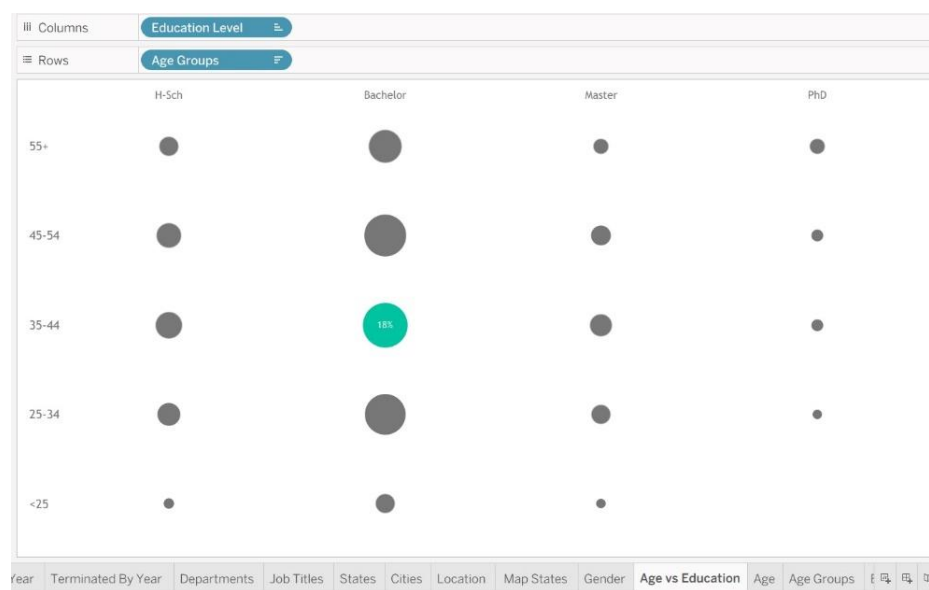
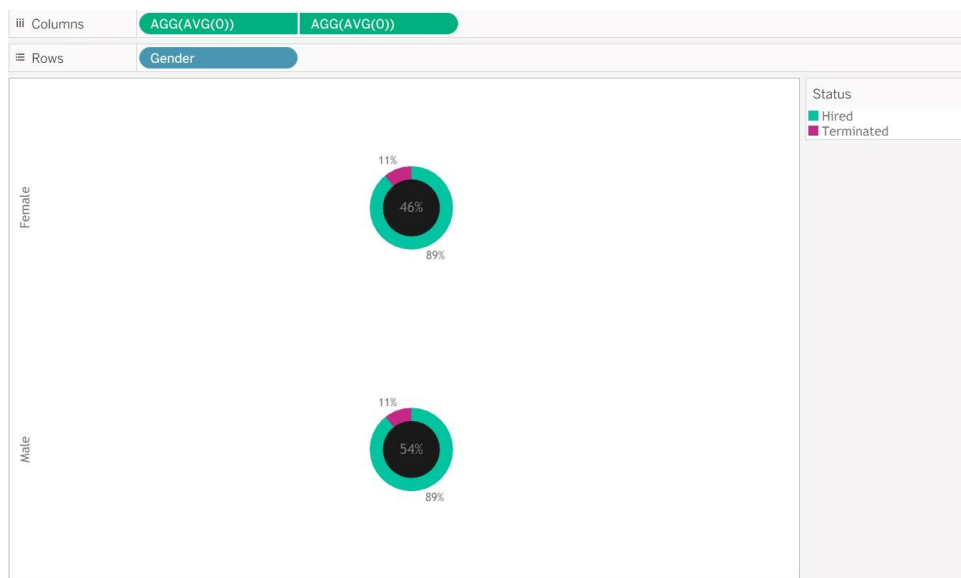
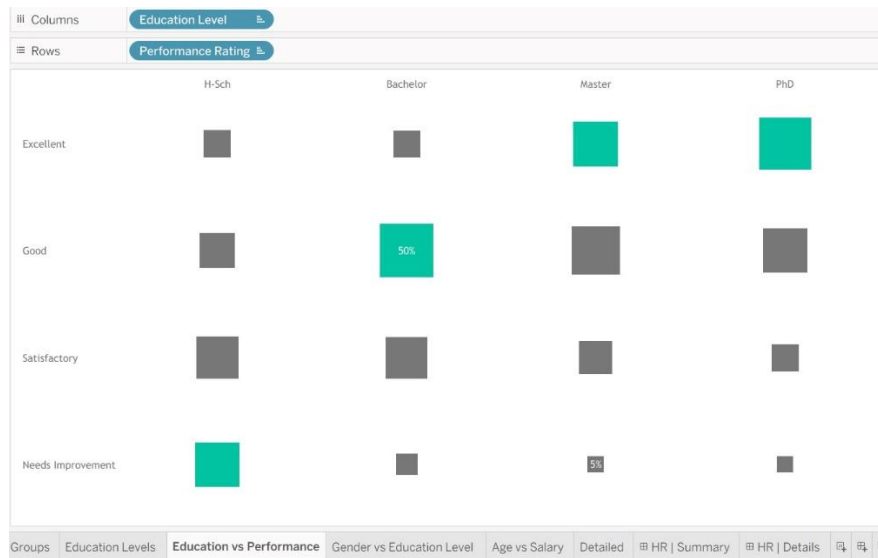
Step 3: Creating Calculated Measures

To support insightful analysis, calculated measures were created within Tableau. Employee age was derived from birthdates, providing a basis for age distribution analysis. Counts of active versus terminated employees were calculated using conditional logic, enabling accurate tracking of workforce trends. Average salaries were computed across groups such as gender, education level, and department to highlight compensation differences. These measures were critical for generating meaningful visualizations that addressed the business questions posed by the HR manager.

- % Highlight Max
- % Total Hired
- % Total Terminated
- Age
- Highlight Max
- Length of Hire
- Rank Top 1
- Rank Top 2
- Salary
- Total Active
- Total Hired
- Total Terminated

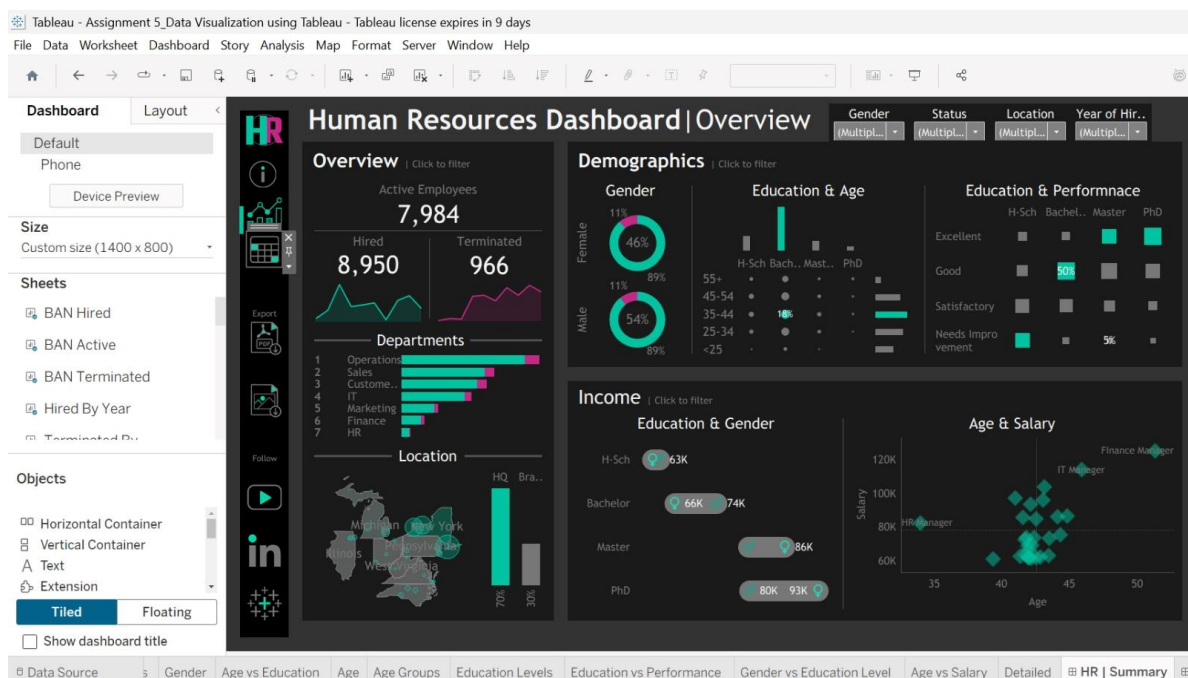
Step 4: Designing Visualizations

Multiple visualization types were created to represent different aspects of the HR dataset. In the Overview section, key metrics were displayed through KPI cards and line charts that illustrated hiring and termination trends over time. Bar charts were used to compare employee counts across departments and job titles, while maps displayed the geographic distribution of employees across states and cities. The Demographics section incorporated charts that highlighted gender ratios, age group distributions, and education levels, providing a clear picture of workforce composition. The Income Analysis section focused on salary comparisons, employing scatter plots to examine the relationship between age and salary within departments, and bar charts to visualize differences by gender and education. Each visualization was designed to be interactive, with filters applied to enable synchronized exploration of the data across multiple dimensions. Below are some of the visualizations.



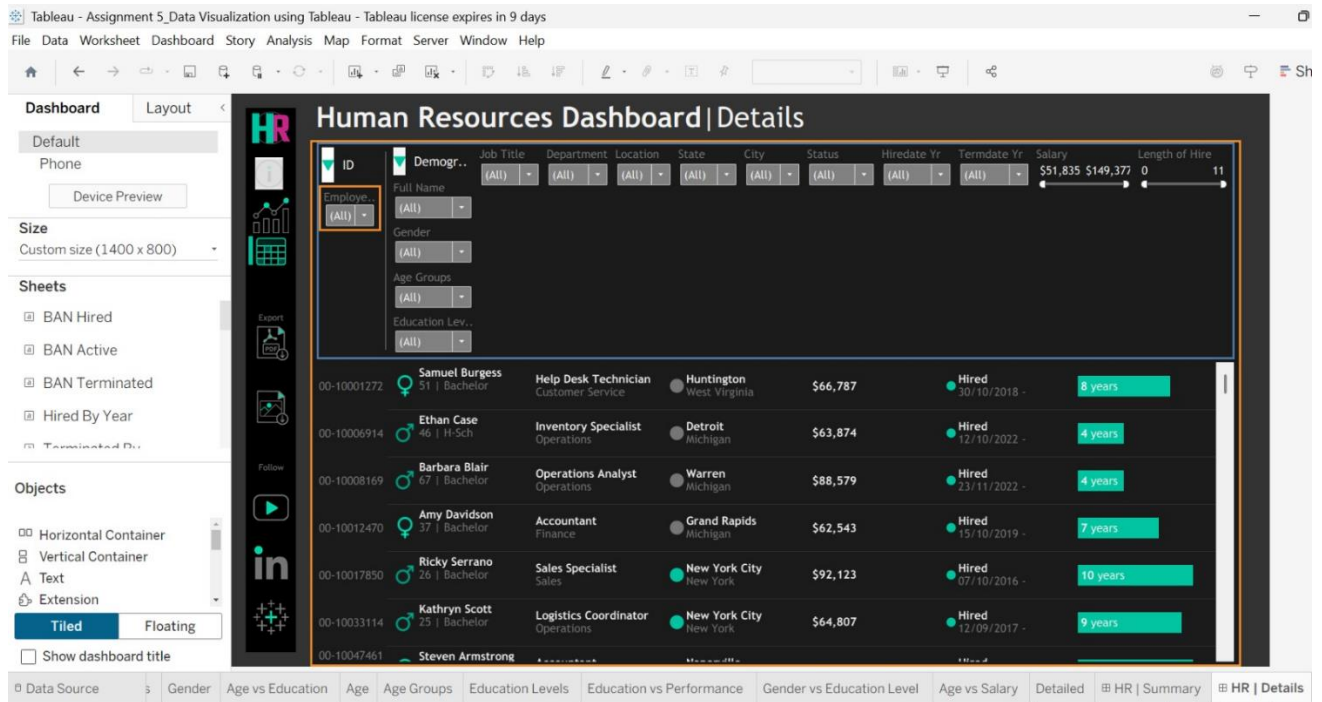
Step 5: Building the Dashboard

All visualizations and calculated measures were combined into a single interactive dashboard, divided into Overview, Demographics, and Income Analysis sections. The layout was designed to be user-friendly, enabling executives and HR personnel to easily navigate between summary insights and detailed data. Interactive filters allowed users to dynamically select departments, genders, or geographic locations, providing the flexibility to explore the workforce data in depth. The dashboard offered a balance between visual appeal, usability, and actionable insights, meeting the specific needs of the HR manager.



Step 6: Detailed Employee Records View

A separate worksheet displayed a comprehensive table containing detailed employee information, including names, departments, positions, gender, age, education, and salary. This table allowed filtering of records based on any column, facilitating quick retrieval of individual profiles or analysis of specific employee subsets.



Link to Project: <https://drive.google.com/file/d/1nsOSg-IMTKo0uVKp4L4QpZpr6XJGY28Q/view>

Conclusion

The process of working on this assignment provided valuable experience in interpreting business requirements, transforming and cleaning data, creating calculated measures, designing multiple interactive visualizations, and assembling them into a coherent dashboard. The final dashboard met the HR manager's needs, offering both high-level insights and detailed employee records that could be explored dynamically. Overall, this project strengthened my skills in Tableau and demonstrated how raw HR data can be converted into actionable and interactive business intelligence tools.